WEEK 4 FINAL PROJECT REPORT

House Price Prediction Using Machine Learning

AI/ML Internship • Week 4 Capstone Project

1. Introduction

This project develops a machine-learning workflow to estimate house prices from housing-property features. A Random Forest Regressor is trained using a housing dataset, and a Flask API is created to serve predictions.

2. Objectives

Explore and preprocess housing data.

Convert categorical features into numerical features.

Train and evaluate a regression model.

Save the trained model for reuse.

Create and test a Flask API for house-price predictions.

3. Dataset Description

The project uses the Housing Dataset loaded in Kaggle. It contains 545 rows and 13 columns. The target variable is `price`; other columns describe property characteristics such as area, bedrooms, bathrooms, stories, parking, and amenities.

4. Tools and Technologies

Python

Pandas and NumPy

Scikit-learn

Random Forest Regressor

Flask

Joblib

5. Methodology

Data loading: Loaded the CSV dataset into a Pandas DataFrame.

Data inspection: Reviewed dataset structure, missing values, duplicate rows, and descriptive statistics.

Preprocessing: Separated `price` as the target and used one-hot encoding for categorical features.

Train/test split: Used an 80:20 split with random_state=42.

Model training: Trained a Random Forest Regressor with 100 estimators and random_state=42.

Evaluation: Calculated MAE, RMSE, and R^2 score.

Model serialization: Saved the trained model as `house_price_model.joblib` and metadata as `model_info.json`.

API development: Created a Flask app with a home endpoint and a `/predict` POST endpoint.

6. Model Evaluation

Actual Model Evaluation Results (Kaggle)

MAE: 1,021,546.04

RMSE: 1,400,565.97

R^2 Score: 0.6119

Interpretation: The model explains approximately 61.19% of the variation in house prices on the test set. MAE and RMSE are reported in the dataset's price units; lower values generally indicate smaller prediction errors.

The model was evaluated using MAE, RMSE, and R^2 score. The exact numeric results should be copied from the Kaggle notebook output before final submission; they are not stated here because the actual values were not available in the conversation.

7. API Development and Testing

The Flask API was started locally and its home endpoint was opened in a browser. A POST request was sent to `/predict`; it returned HTTP status code 200 and a predicted-price JSON response. The model-loading output showed a scikit-learn version mismatch warning (model saved with version 1.6.1, local environment using 1.9.1). Although the API responded, matching library versions is recommended before relying on the predictions.

8. Project Files

`app.py` — Flask API implementation.

`house_price_model.joblib` — serialized trained model.

`model_info.json` — model metadata and evaluation details.

`requirements.txt` — Python dependencies.

`README.md` — project documentation.

`house_price_project.zip` — bundled project files.

9. Conclusion

The project demonstrates an end-to-end machine-learning workflow, from data preprocessing and model training to model serialization and API testing. The local API was tested successfully with an HTTP 200 response.

10. Future Improvements

Align the Scikit-learn version used for training and local inference.

Validate required API fields and categorical values.

Add automated tests and clearer API error handling.

Tune model hyperparameters and document final evaluation metrics.

Deploy the application to a suitable hosting platform.

Before submitting: Insert the exact MAE, RMSE, and R^2 values from the Kaggle evaluation output.

Metric	Purpose
MAE	Average absolute difference between actual and predicted prices.
RMSE	Error metric that penalizes larger errors more strongly.
R^2 Score	Measures how much variation in the target is explained by the model.